MODELING GREAT SALT LAKE DUST AND AIR QUALITY

Recommended direction for a physics-informed, data-driven model

Core question

How much does dust emitted from exposed Great Salt Lake lakebed contribute to PM10 concentrations at Wasatch Front monitoring sites, and how does that contribution change with wind, atmospheric mixing, and lake level?

Prepared September 2026

1. Recommended modeling direction

> **Recommendation: build a simple physics-based state-space model first, compare it against a strong machine-learning baseline, and then use a hybrid model in which ML learns the residual error left by the physics model.**

The project should not initially try to predict "air quality" in general. The more defensible target is the incremental coarse-particle signal associated with Great Salt Lake dust, with PM10 as the primary response variable. That makes the source mechanism explicit and reduces the risk that the model simply learns traffic, wildfire smoke, winter inversions, or other unrelated pollution patterns.

A good first version uses one monitoring site, hourly data, and a limited set of interpretable drivers: current PM10, wind speed and direction, precipitation or soil wetness, atmospheric mixing height, and lake elevation/exposed playa. Once that works, add more sites and spatial transport.

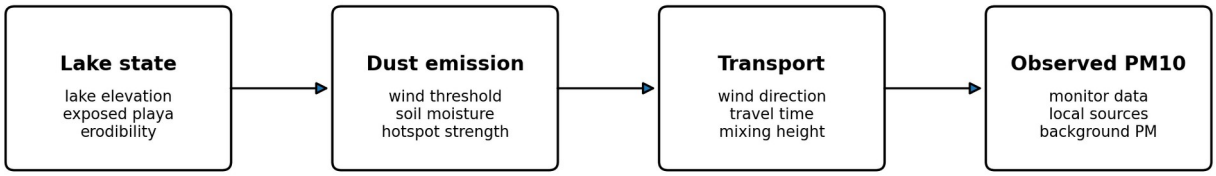


Figure 1. The modeling problem naturally separates into lake state, emission, transport, and the concentration observed at a monitor.

2. What to keep - and change - from the current equation

The proposed equation is a useful conceptual starting point because it treats pollution as a dynamic state rather than a static regression target:

$$P'(t) = (alpha + P(t))\, S(t) + kappa + f(t)$$

Current conceptual form

The structure needs modification before fitting data. The main issues are:

- Wind speed alone is not a source term. Wind can transport lake dust toward a monitor, move it away from the monitor, or dilute the local air mass.
- The equation mixes background pollution, lake emissions, and transport into terms that will be difficult to identify separately from observations.
- There is no explicit removal mechanism for deposition, ventilation, or precipitation scavenging.
- The units need to be consistent. For particulate matter, use mass concentration such as micrograms per cubic meter rather than ppm.
- The Great Salt Lake source should depend on whether the exposed playa is actually emitting dust, which usually requires sufficiently strong winds and dry/erodible surface conditions.

The part worth preserving is the ODE/state-space viewpoint: pollution today depends on pollution a moment ago plus physically meaningful source, transport, and loss terms.

3. A better first model

For a monitoring site i, start with a discrete-time version of a simple mass-balance model. Hourly time steps are practical because both particulate and weather data are commonly available at that resolution.

$$P[t+1] = P[t] + dt * \{ k_v(t) [P_bg(t) - P[t]] + E_GSL(t)/H(t) + E_local(t) - lambda\ P[t] \}$$

Recommended first-order state equation

Interpretation:

- P[t]: PM10 concentration at the target monitor.
- P_bg(t): background particulate concentration entering the region.
- k_v(t): ventilation/advection rate. Faster atmospheric exchange pulls local concentration toward the upwind background state.
- E_GSL(t): Great Salt Lake dust contribution reaching the local air column.
- H(t): boundary-layer or mixing height. The same emitted dust produces a lower near-surface concentration when mixed through a deeper atmosphere.
- E_local(t): local/non-lake sources such as construction, roads, mining, or other dust.
- lambda: effective removal rate from deposition and other losses.

The key modeling work is the lake source term. A tractable first parameterization is:

$$E_GSL(t) = beta * A_eff(t) * R(t) * max[U(t) - U_c, 0]^n * D_i(t)$$

A simple threshold dust-emission and transport term

Here A_eff is exposed/available source area, R is an optional erodibility factor, U_c is a threshold wind speed, n controls how sharply emissions increase above the threshold, and D_i is a directional transport factor between the lake and the monitor. For a first pass, D_i can be based on wind direction and distance. A later version can replace it with trajectory or gridded wind information.

One useful refinement is a travel-time lag. If a source area is d kilometers from the monitor and the relevant wind speed is U, then use a lagged source E_GSL(t - tau), with tau approximately d/U. This prevents the model from pretending dust emitted over the lake is observed instantaneously in Salt Lake City.

4. Data to assemble

The data situation is unusually favorable. The following sources are sufficient for a strong first project, and newer Utah dust-monitoring infrastructure can support later source-attribution work.

Data	Primary source	Useful variables	Role in model
Air quality	EPA AQS / Utah DAQ	Hourly PM10, PM2.5, PMc; monitor metadata	Response variable and event detection
Surface weather	MesoWest / Synoptic	Wind speed, direction, gusts, RH, temperature, precipitation	Emission and transport drivers
Atmospheric structure	NOAA HRRR	Boundary-layer height, gridded winds, precipitation	Mixing and spatial transport
Lake state	USGS	Water-surface elevation; elevation-area-volume tables	Estimate exposed lakebed area
Dust-specific monitoring	Utah DEQ UDORN	PM10, event records, filter chemistry, meteorology	Source attribution and validation
Hotspots / prior models	University of Utah / Wilkes Center	Mapped dust hotspots and GSL dust exposure model	Benchmark assumptions and spatial source strength

Important distinction: predicting PM10 is easier than proving the Great Salt Lake caused a particular PM10 spike. Source attribution becomes much stronger when wind trajectories, upwind source geometry, and eventually dust chemistry agree with one another.

5. The model ladder: what to compare

Do not judge the physics model in isolation. Build a small hierarchy of models and compare them on the same out-of-sample periods.

1. Persistence baseline. Predict P[t+1] = P[t]. Air pollution is autocorrelated, so any useful model should beat this simple benchmark.
2. Meteorology-only statistical baseline. Use linear regression or a generalized additive model with wind, direction, humidity, precipitation, hour, month, and recent PM10.
3. Pure ML baseline. XGBoost or LightGBM is appropriate. Deep learning is unnecessary at this stage. The ML model answers: how predictable is PM10 from the available features without imposing physical structure?
4. Physics-informed state model. Fit beta, U_c, n, lambda, ventilation terms, and any site-specific coefficients by minimizing forecast error subject to sensible parameter bounds.
5. Hybrid residual model. Let the physics model make the first prediction, then train ML on its residuals: prediction = physics + learned correction. This often gives the best balance of interpretability and predictive power.

Why this is better than "just ML": the physics model gives parameters and counterfactuals you can interpret. The ML baseline prevents you from congratulating a mechanistic model that performs worse than a generic predictor.

6. Treat dust events explicitly

A useful complementary formulation is a two-stage model: first estimate whether a Great Salt Lake dust event is occurring, then estimate its magnitude. This is especially useful because dust is episodic rather than a continuous source at the same strength every hour.

$$Pr(\text{event at } t) = g(\text{wind, direction, lake state, dryness, precipitation, mixing, season})$$

Stage 1: event probability

$$\Delta PM10(t) = h(\text{features}) \mid \text{event}$$

Stage 2: conditional concentration enhancement

The event model can be used for validation even if the continuous ODE remains the main scientific model. For example, the inferred lake contribution should be near zero when winds are not transporting air from exposed lakebed toward the monitor.

7. Evaluation strategy

- Use time-based train/validation/test splits. Do not randomly shuffle hours; temporal autocorrelation would leak information across the split.
- Hold out entire dust events, not only individual hours, so the model must generalize to new storms.
- Report MAE/RMSE for ordinary hours plus event-focused metrics such as peak error, peak timing, and recall of high-PM10 episodes.
- Evaluate separately during wildfire smoke, winter inversions, and strong regional dust events. These are major confounders.
- Check physical sanity: lake contribution should fall when the lake is not upwind, after precipitation, or when winds remain below the emission threshold.
- Use uncertainty intervals or bootstrap estimates for fitted physical parameters. A point estimate of U_c or beta without uncertainty can look more precise than the data justify.

For causal language, be conservative. A predictive model can estimate a Great Salt Lake contribution under its assumptions, but strong causal attribution requires evidence that separates lake dust from Sevier Lake, West Desert sources, mining, roads, construction, and wildfire smoke. Chemistry and trajectory data are especially valuable here.

8. Minimum viable project

A project that is both finishable and scientifically useful would look like this:

Choose one primary PM10 monitor. Prefer a site with a long continuous record and meaningful Great Salt Lake exposure. Hawthorne or the DEQ Tech Center are reasonable candidates for historical work; a near-lake UDORN site can become a second-stage validation site. Verify actual data completeness before locking the site.

Build an hourly analysis table. Merge PM10, wind speed/direction, gusts, temperature, humidity, precipitation, lake elevation, estimated exposed area, mixing height, and lagged PM10. Add hour-of-day and season features only as controls, not as substitutes for physics.

Plot and label 20-50 major episodes. For each high-PM10 event, plot PM10, wind direction, wind speed, precipitation, and lake-to-monitor geometry. This manual review will expose data problems and clarify which events are plausibly lake-related.

Fit the simple physics model. Estimate the wind threshold, emission exponent, source-strength coefficient, ventilation/removal terms, and any lag. Keep the number of free parameters small.

Fit XGBoost on the same features. Use exactly the same training period and test events. Add SHAP or feature importance only after predictive performance is established.

Fit a hybrid residual model. Train the ML model to predict the physics model residual. Test whether the residual learner improves ordinary hours without destroying performance during dust events.

Run lake-level counterfactuals. Hold the observed weather fixed and vary the lake-elevation/exposed-area input. This produces a transparent scenario analysis of how changing lake state alters predicted dust exposure under the model.

9. Extensions after the first model works

- Multi-site spatial model: fit several monitors jointly and let source contribution decay or disperse with distance.
- Back-trajectory model: use HYSPLIT or HRRR winds to estimate whether observed air parcels crossed exposed lakebed before reaching each sensor.
- Hotspot weighting: replace total exposed area with mapped erodible/hotspot area so newly exposed but crusted surfaces do not count equally.
- Chemical source attribution: use filter chemistry or heavy-metal fingerprints to distinguish Great Salt Lake playa dust from other regional and anthropogenic sources.
- Bay-specific sources: separate Farmington Bay, Bear River Bay, Stansbury Bay, and other source regions rather than treating the lake as one homogeneous emitter.
- Bayesian state-space model: once the simple model is understood, place priors on physical parameters and propagate uncertainty through the inferred lake contribution.

10. Bottom line

> **Build the interpretable model first. The strongest version of this project is not "use ML to predict Salt Lake air quality." It is "construct a physically interpretable model of Great Salt Lake dust emission and transport, test it against observed PM10, and use ML as a benchmark and residual corrector."**

That structure gives you three things at once: a predictive model, estimates of physically meaningful parameters, and a framework for counterfactual lake-level scenarios. It is also naturally extensible as UDORN and other dust-specific datasets become richer.

11. A practical six-week build sequence

Week 1: Pull one monitor of hourly PM10 plus nearby weather; clean timestamps, missing values, and units.

Week 2: Merge lake elevation/exposed-area data and HRRR mixing height. Build event plots and identify plausible lake-dust episodes.

Week 3: Fit persistence, linear/GAM, and XGBoost baselines using a time-based split.

Week 4: Implement and fit the physics state model. Stress-test wind direction, lag, threshold wind speed, and removal assumptions.

Week 5: Train the residual ML model and evaluate on held-out storms, wildfire periods, and ordinary days.

Week 6: Run lake-level counterfactuals, quantify parameter uncertainty, and write up where the model succeeds and fails.

Sources and starting points

1. Utah DEQ - Dust Monitoring Network (UDORN). https://deq.utah.gov/air-quality/dust-monitoring-network Network design, station roles, PM10 monitoring, filter chemistry, and 2026-2031 operating plan.

2. Utah DEQ - Current Dust Research Projects. https://deq.utah.gov/air-quality/current-dust-research-projects UDORN, the 25-year Dust Event Legacy Database, and heavy-metals inventory.

3. US EPA - AQS AirData Downloads. https://aqs.epa.gov/aqsweb/airdata/download_files.html Historical hourly PM10, PM2.5, PMc and other ambient air-quality data.

4. USGS - Great Salt Lake Elevations and Areal Extent. https://www.usgs.gov/centers/utah-water-science-center/science/great-salt-lake-elevations-and-areal-extent Long lake-elevation record and links to topobathymetry / area-volume relationships.

5. USGS - Great Salt Lake Elevation-Area-Volume Tables. https://data.usgs.gov/datacatalog/data/USGS%3A6467b42fd34ec11ae4a8afb1 CSV tables mapping lake elevation to area and volume.

6. MesoWest User Guide. https://mesowest.utah.edu/html/help/userguide.html Archived surface weather observations including wind, temperature, humidity, and precipitation.

7. NOAA - High-Resolution Rapid Refresh (HRRR). https://rapidrefresh.noaa.gov/hrrr/ Hourly 3-km atmospheric model useful for gridded winds and mixing structure.

8. University of Utah - GSL Dust Exposure Model. https://science.utah.edu/faculty/faculty-research/map-tool-gsldust/ Existing data-based Great Salt Lake dust exposure tool by Derek Mallia and Kevin Perry; useful benchmark for assumptions.

9. University of Utah - Mapping the Great Salt Lake Dust Hotspots. https://science.utah.edu/faculty/faculty-research/mapping-gsl-hotspots/ Background on spatially heterogeneous dust-producing areas of the exposed lakebed.

Recommended first implementation stack: Python + pandas/xarray for data, scipy/statsmodels for parameter fitting, scikit-learn/XGBoost for baselines, and matplotlib for event diagnostics. Add trajectory tooling only after the one-site model is stable.